

DISCRETE MATHEMATICS – CONTEST 4

Complete Exam Revision Notes

Part 2 (Abstract Algebra + Advanced Counting): 60–70% focus • Part 1 (Logic, Proofs, Sets): 30–40% focus

How to Use These Notes: Sections marked **[HIGH PRIORITY]** are Part 2 topics – study these first and most thoroughly. Sections marked **[PART 1]** are important but covered more concisely.

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PART 1 – Logic, Proofs, Set Theory (30–40 percent of exam)

Cover these topics after you are comfortable with Part 2. Know the key definitions, main theorems, and be able to do basic examples.

[PART 1] Logic and Its Applications**1.1 Propositional Logic****Core Connectives**

Let p, q be propositions (statements that are TRUE or FALSE).

Name	Symbol	Read as	True when
Negation	$\neg p$	“not p ”	p is false
Conjunction	$p \wedge q$	“ p and q ”	both true
Disjunction	$p \vee q$	“ p or q ”	at least one true
Implication	$p \rightarrow q$	“if p then q ”	not (p true, q false)
Biconditional	$p \leftrightarrow q$	“ p iff q ”	same truth value

Key Logical Equivalences (Memorise These!)**De Morgan’s Laws:**

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Implication:

$$p \rightarrow q \equiv \neg p \vee q$$

Contrapositive (same as original):

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

Converse (NOT equivalent):

$$q \rightarrow p \text{ (different!)}$$

Double Negation:

$$\neg(\neg p) \equiv p$$

Absorption:

$$p \wedge (p \vee q) \equiv p$$

Quick Check – Truth Table for p implies q

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

The only way an implication is false: premise true, conclusion false.

1.2 Predicate Logic & Quantifiers**Quantifiers**

- **Universal:** $\forall x P(x)$ — “For all x , $P(x)$ is true.” False if even one counterexample exists.
- **Existential:** $\exists x P(x)$ — “There exists an x such that $P(x)$.” True if at least one witness exists.

Negation of Quantifiers (De Morgan for predicates):

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

Nested: $\forall x \exists y P(x, y)$ means: for every x , there is *some* y (may depend on x).

Order matters: $\forall x \exists y \neq \exists y \forall x$ in general.

1.3 Boolean Algebra (Computer Architecture)**Boolean Algebra Laws**

$\wedge \leftrightarrow$ multiplication (AND), $\vee \leftrightarrow$ addition (OR), $\neg \leftrightarrow$ complement (NOT). Key: $x \wedge \bar{x} = 0$, $x \vee \bar{x} = 1$, $x \wedge x = x$, $x \vee x = x$ (idempotent).

SOP (Sum of Products) and **POS** (Product of Sums) are standard forms for logic circuits.

[PART 1] Proofs**2.2 Rules of Inference****Key Inference Rules****Modus Ponens:**

$$\frac{p, \quad p \rightarrow q}{q}$$

Modus Tollens:

$$\frac{\neg q, \quad p \rightarrow q}{\neg p}$$

Hypothetical Syllogism:

$$\frac{p \rightarrow q, \quad q \rightarrow r}{p \rightarrow r}$$

Disjunctive Syllogism:

$$\frac{p \vee q, \quad \neg p}{q}$$

2.3 Types of Proofs**Proof Types – Know These!**

Direct Proof: Assume p , show q follows logically.

Proof by Contrapositive: Prove $\neg q \rightarrow \neg p$ instead of $p \rightarrow q$.

Proof by Contradiction: Assume $\neg p$, derive a contradiction $\perp$.

Proof by Induction: Base case $P(1)$; Inductive step: $P(k) \rightarrow P(k+1)$.

Proof by Cases: Split into exhaustive cases and prove each.

Classic Example – $\sqrt{2}$ is irrational (Contradiction)

Assume $\sqrt{2} = p/q$ in lowest terms. Then $2q^2 = p^2 \Rightarrow p^2$ even $\Rightarrow p$ even $\Rightarrow p = 2k \Rightarrow 2q^2 = 4k^2 \Rightarrow q^2 = 2k^2 \Rightarrow q$ even. But p, q both even contradicts lowest terms. $\square$

2.4 Loop Invariants

Loop Invariant – Quick Concept

A **loop invariant** is a property that holds *before, during, and after* each iteration. To prove program correctness:

1. **Initialization:** Invariant holds before loop starts.
2. **Maintenance:** If invariant holds at start of iteration, it holds at end.
3. **Termination:** When loop ends, invariant + termination condition $\Rightarrow$ correctness.

[PART 1] Set Theory

3.1 Russell's Paradox

Russell's Paradox

Let $S = \{x : x \notin x\}$. Is $S \in S$?

$S \in S \Rightarrow S \notin S$ (by definition). $S \notin S \Rightarrow S \in S$. Contradiction!

This shows naive set comprehension is inconsistent; motivates axiomatic set theory (ZFC).

3.2 & 3.3 Countability and Uncountability

Countability

A set A is **countable** if there is a bijection $f : \mathbb{N} \rightarrow A$ (or A is finite).

Countable: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{N} \times \mathbb{N}$, any finite union of countable sets.

Uncountable: $\mathbb{R}$, $[0, 1]$, $\mathcal{P}(\mathbb{N})$ (power set of naturals).

Cantor Diagonal Argument – $\mathbb{R}$ is Uncountable

Suppose all reals in $[0, 1]$ are listed: $r_1, r_2, r_3, \dots$. Build d where $d_i \neq$ the i -th digit of r_i . Then d differs from every r_i in at least one digit, so d is not in the list. Contradiction. Therefore $[0, 1]$ is uncountable.

Application in Computability

The set of all programs is countable (strings over finite alphabet). The set of all functions $\mathbb{N} \rightarrow \mathbb{N}$ is uncountable. Therefore, **most functions cannot be computed** by any program — the Halting Problem is undecidable.

PART 2 – Abstract Algebra & Advanced Counting (60–70 percent of exam – STUDY THIS FIRST)

These are the most important topics. Know all definitions, theorems, and be able to work examples by hand. This section is detailed and exam-focused.

[HIGH PRIORITY] Abstract Algebra

4.1 Axioms of Group Theory

Definition of a Group

A **group** $(G, \star)$ is a set G with a binary operation $\star$ satisfying:

1. **Closure:** $\forall a, b \in G, a \star b \in G$
2. **Associativity:** $\forall a, b, c \in G, (a \star b) \star c = a \star (b \star c)$
3. **Identity:** $\exists e \in G$ such that $\forall a \in G, e \star a = a \star e = a$
4. **Inverse:** $\forall a \in G, \exists a^{-1} \in G$ such that $a \star a^{-1} = a^{-1} \star a = e$

If additionally $a \star b = b \star a$ for all a, b , then G is **abelian** (commutative).

Memory Trick: CAII

Closure, Associativity, Identity, Inverse $\Rightarrow$ GROUP. Add commutativity $\Rightarrow$ Abelian group.

Examples of Groups

- $(\mathbb{Z}, +)$: integers under addition. Identity = 0. Inverse of a is $-a$. **Abelian.**
- $(\mathbb{R} \setminus \{0\}, \times)$: nonzero reals under multiplication. Identity = 1. Inverse of a is $1/a$. **Abelian.**
- $(\mathbb{Z}_n, +_n)$: integers mod n under addition. **Abelian.**
- S_n (symmetric group, permutations): **Non-abelian** for $n \geq 3$.
- $(\mathbb{Z}_n^*, \times_n)$: units mod n under multiplication.

4.2 & 4.3 Addition and Multiplication Modulo Groups

Modular Arithmetic

For $n \in \mathbb{Z}^+$, define $a \equiv b \pmod{n}$ if $n \mid (a - b)$.

- $a + b \pmod{n}$: add then take remainder.
- $a \times b \pmod{n}$: multiply then take remainder.

$(\mathbb{Z}_n, +_n) = \{0, 1, 2, \dots, n-1\}$ under addition mod n . This is always a group.

$(\mathbb{Z}_n^*, \times_n)$: elements of $\{1, \dots, n-1\}$ with $\gcd(a, n) = 1$. This is a group under multiplication. Its order is $\phi(n)$ (Euler's totient function).

Euler's Theorem and Fermat's Little Theorem

Euler's Theorem: If $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Fermat's Little Theorem: If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.
(Fermat is a special case of Euler when $n = p$ prime, since $\phi(p) = p - 1$.)

Euler's Totient: $\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$.

For prime p : $\phi(p) = p - 1$. For $n = p^k$: $\phi(p^k) = p^{k-1}(p - 1)$.

Example: $\mathbb{Z}_6$ Addition and $\mathbb{Z}_7^*$ Multiplication

$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ under $+_6$: every element has an inverse ($0 \leftrightarrow 0, 1 \leftrightarrow 5, 2 \leftrightarrow 4, 3 \leftrightarrow 3$).

$\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$ under $\times_7$. Since 7 is prime, $\phi(7) = 6$, so $|\mathbb{Z}_7^*| = 6$. Example: $3^{-1} \pmod{7} = 5$ because $3 \times 5 = 15 \equiv 1$.

4.5 Order of an Element and Order of a Group

Order Definitions

- **Order of a group G :** $|G|$ = number of elements in G .
- **Order of an element $a \in G$:** The smallest positive integer k such that $a^k = e$ (identity). Denoted $\text{ord}(a)$ or $|a|$.
- If no such k exists, a has **infinite order**.

Key Theorem on Order of Elements

In a finite group G with $|G| = n$, for any $a \in G$:

- $a^n = e$ (the identity)
- $\text{ord}(a)$ divides $|G|$ (a consequence of Lagrange's Theorem)
- $a^k = e \iff \text{ord}(a) \mid k$

Example: Finding Orders in $\mathbb{Z}_8$

In $(\mathbb{Z}_8, +)$: identity = 0, operation is addition.

$\text{ord}(1)$: 1, 2, 3, 4, 5, 6, 7, 0 — takes 8 steps. $\text{ord}(1) = 8$.

$\text{ord}(2)$: 2, 4, 6, 0 — takes 4 steps. $\text{ord}(2) = 4$.

$\text{ord}(4)$: 4, 0 — takes 2 steps. $\text{ord}(4) = 2$.

$\text{ord}(0) = 1$ (identity).

All orders $\{1, 2, 4, 8\}$ divide $|G| = 8$. ✓

4.6 Cyclic Groups

Cyclic Groups

A group G is **cyclic** if there exists $g \in G$ such that every element of G is a power of g :

$$G = \langle g \rangle = \{g^0 = e, g^1, g^2, \dots, g^{n-1}\}$$

g is called a **generator** of G .

- Every cyclic group is **abelian**.

- $\mathbb{Z}_n = \langle 1 \rangle$ is cyclic of order n .
- $\mathbb{Z}$ is infinite cyclic, generated by 1 (or -1).

Theorems on Cyclic Groups

Theorem 1: $a \in \mathbb{Z}_n$ is a generator of $\mathbb{Z}_n$ iff $\gcd(a, n) = 1$.

Theorem 2: The number of generators of $\mathbb{Z}_n$ is $\phi(n)$.

Theorem 3: Every subgroup of a cyclic group is cyclic.

Theorem 4: A cyclic group of order n has exactly one subgroup of order d for every divisor d of n .

Theorem 5: In $\langle g \rangle$ of order n : $\text{ord}(g^k) = \frac{n}{\gcd(n, k)}$.

Example: Generators of $\mathbb{Z}_{12}$

$\phi(12) = 12 \cdot (1 - \frac{1}{2})(1 - \frac{1}{3}) = 4$. Generators are a with $\gcd(a, 12) = 1$: $\{1, 5, 7, 11\}$.
Check: $\langle 5 \rangle$: 5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, 0 — hits all 12 elements. ✓

Example: Primitive Root (Generator of $\mathbb{Z}_p^*$)

In $\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$, is 3 a generator?
 $3^1 = 3, 3^2 = 2, 3^3 = 6, 3^4 = 4, 3^5 = 5, 3^6 = 1 \pmod{7}$. Yes! 3 generates all of $\mathbb{Z}_7^*$, so 3 is a **primitive root mod 7**.

4.7 Lagrange's Theorem and Corollaries

Lagrange Theorem – VERY IMPORTANT

Theorem: Let H be a subgroup of a finite group G . Then $|H|$ divides $|G|$.
More precisely: $|G| = |H| \cdot [G : H]$ where $[G : H]$ is the number of distinct (left) cosets of H in G .

Proof sketch: The left cosets $\{aH : a \in G\}$ partition G , and each coset has $|H|$ elements. So $|G| = [G : H] \cdot |H|$.

Cosets

For $H \leq G$ and $a \in G$:

- **Left coset:** $aH = \{ah : h \in H\}$
- **Right coset:** $Ha = \{ha : h \in H\}$

Two cosets are either equal or disjoint. They form a partition of G .
If G is abelian, left and right cosets coincide ($aH = Ha$).

Corollaries of Lagrange's Theorem

Corollary 1: The order of any element $a \in G$ divides $|G|$.
(Because $\langle a \rangle$ is a subgroup of order $\text{ord}(a)$.)

Corollary 2: For any $a \in G$: $a^{|G|} = e$.

Corollary 3: If $|G|$ is prime, then G is cyclic and every non-identity element is a generator.

Corollary 4 (Fermat's Little Theorem from group theory): For prime p and $a \not\equiv 0$:
 $a^{p-1} \equiv 1 \pmod{p}$.

Example: Cosets of $H=\{0,3\}$ in $\mathbb{Z}_6$

$H = \{0, 3\} \leq \mathbb{Z}_6$ ($|H| = 2$ divides $|\mathbb{Z}_6| = 6$, $\checkmark$). Left cosets:
 $0 + H = \{0, 3\}$, $1 + H = \{1, 4\}$, $2 + H = \{2, 5\}$.
 Three cosets, each of size 2, partitioning $\mathbb{Z}_6$. $[G : H] = 3$.

4.8 Axioms of Ring Theory

Definition of a Ring

A **ring** $(R, +, \times)$ is a set R with two operations satisfying:

1. $(R, +)$ is an **abelian group** (with identity 0 and inverses $-a$)
2. Multiplication is **associative**: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
3. **Distributive laws**: $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$

Additional properties (not required for a ring):

- If $\cdot$ is commutative: **commutative ring**
- If $\exists 1$ (multiplicative identity): **ring with unity**
- If every nonzero element has a multiplicative inverse: **division ring**
- Commutative ring with unity, no zero divisors: **integral domain**

Examples of Rings

- $(\mathbb{Z}, +, \times)$: commutative ring with unity 1. Integral domain.
- $(\mathbb{Z}_n, +_n, \times_n)$: commutative ring. Is an integral domain iff n is prime (then it's a field).
- $(\mathbb{Z}[x], +, \times)$: polynomial ring over integers.
- $M_n(\mathbb{R})$: $n \times n$ matrices – non-commutative ring with unity ($n \geq 2$).

4.9 Polynomial Rings

Polynomial Ring $R[x]$

For a ring R , the **polynomial ring** $R[x]$ consists of all polynomials $a_n x^n + \dots + a_1 x + a_0$ with $a_i \in R$.

Operations: pointwise addition, standard polynomial multiplication.

Degree: $\deg(f)$ = highest power with nonzero coefficient. $\deg(fg) = \deg(f) + \deg(g)$ (if R is integral domain).

$F[x]$ **for a field F** : Division algorithm holds: for $f, g \in F[x]$, $g \neq 0$:

$$f = q \cdot g + r, \quad \deg(r) < \deg(g)$$

$F[x]$ is a **Euclidean domain** $\Rightarrow$ gcds exist, unique factorisation holds.

Irreducible Polynomials

$p(x) \in F[x]$ is **irreducible** over F if $\deg(p) \geq 1$ and it cannot be written as a product of two polynomials of lower degree in $F[x]$.

Example: $x^2 + 1$ is irreducible over $\mathbb{R}$ but factors as $(x + i)(x - i)$ over $\mathbb{C}$.
 $x^2 + x + 1$ is irreducible over $\mathbb{Z}_2$ (check: $f(0) = 1 \neq 0, f(1) = 1 \neq 0$).

4.10 Axioms of Fields

Definition of a Field

A **field** $(F, +, \times)$ is a commutative ring with unity where every nonzero element has a multiplicative inverse. Equivalently:

1. $(F, +)$ is an abelian group (identity: 0)
2. $(F \setminus \{0\}, \times)$ is an abelian group (identity: 1)
3. Distributive law: $a(b + c) = ab + ac$

Examples: $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_p$ (for prime p), $\text{GF}(p^n)$ (finite fields).

Finite Fields – Key Facts

Theorem: A finite field has order p^n for some prime p and $n \geq 1$.

Notation: $\text{GF}(q)$ or $\mathbb{F}_q$ is the unique (up to isomorphism) field of order $q = p^n$.

Construction: $\text{GF}(p^n) \cong \mathbb{Z}_p[x]/\langle f(x) \rangle$ where f is an irreducible polynomial of degree n over $\mathbb{Z}_p$.

Example: $\text{GF}(4) = \mathbb{Z}_2[x]/\langle x^2 + x + 1 \rangle = \{0, 1, x, x + 1\}$ with arithmetic mod $x^2 + x + 1$.

Field vs Ring vs Group Hierarchy

Group $\subset$ Ring $\subset$ Integral Domain $\subset$ Field

Structure	+ group	$\times$ assoc	$\times$ inverses
Group	✓ (just one op)	—	—
Ring	✓ abelian	✓	$\times$
Integral Domain	✓ abelian	✓	$\times$ (no zero div.)
Field	✓ abelian	✓	✓ (nonzero)

4.11 Primitive Elements

Primitive Element (Primitive Root)

An element $g \in \mathbb{F}_q^*$ is a **primitive element** if it is a generator of the multiplicative group $(\mathbb{F}_q^*, \times)$, i.e., $\mathbb{F}_q^* = \langle g \rangle$.

Since $|\mathbb{F}_q^*| = q - 1$, g is primitive iff $\text{ord}(g) = q - 1$.

Theorem: $(\mathbb{F}_q^*, \times)$ is cyclic, so primitive elements always exist. There are $\phi(q - 1)$ of them.

For $\mathbb{Z}_p^*$: g is a primitive root mod p if $g^k \pmod{p}$ cycles through all of $\{1, 2, \dots, p - 1\}$.

Example: Primitive Root mod 5

$\mathbb{Z}_5^* = \{1, 2, 3, 4\}$, $|\mathbb{Z}_5^*| = 4 = \phi(5)$.

Try $g = 2$: $2^1 = 2$, $2^2 = 4$, $2^3 = 3$, $2^4 = 1 \pmod{5}$. Generates $\{1, 2, 3, 4\}$. ✓ So 2 is primitive.

Try $g = 4$: $4^1 = 4$, $4^2 = 1 \pmod{5}$. Order 2. Not primitive.

Number of primitive roots $= \phi(4) = 2$. They are $\{2, 3\}$.

4.12 Cryptography – Discrete Logarithm & Diffie-Hellman**Discrete Logarithm Problem (DLP)**

Given a prime p , a primitive root $g \pmod{p}$, and $h \in \mathbb{Z}_p^*$:

Find integer x such that $g^x \equiv h \pmod{p}$.

$x = \log_g h \pmod{p-1}$ is the **discrete logarithm** of h base g .

Why it's hard: Computing $g^x \pmod{p}$ is easy (fast exponentiation). But finding x given g^x is computationally infeasible for large p — this asymmetry is the security basis.

Diffie-Hellman Key Exchange – CRITICAL FOR EXAM

Setup (public): Large prime p , primitive root $g \pmod{p}$.

Protocol:

1. **Alice** chooses secret a , computes $A = g^a \pmod{p}$, sends A to Bob.
2. **Bob** chooses secret b , computes $B = g^b \pmod{p}$, sends B to Alice.
3. **Alice** computes shared key $K = B^a = g^{ab} \pmod{p}$.
4. **Bob** computes shared key $K = A^b = g^{ab} \pmod{p}$.

Both arrive at the same key $K = g^{ab} \pmod{p}$ without ever transmitting a , b , or K .

Security: Eavesdropper knows $g, p, A = g^a, B = g^b$ but cannot easily find $K = g^{ab}$ — requires solving DLP.

Worked Example: Diffie-Hellman

Public: $p = 23$, $g = 5$ (primitive root mod 23).

Alice: $a = 6$, $A = 5^6 = 15625 \equiv 8 \pmod{23}$. Sends $A = 8$.

Bob: $b = 15$, $B = 5^{15} \pmod{23}$. $5^{15} = 5^8 \cdot 5^4 \cdot 5^2 \cdot 5^1$. We get $B \equiv 19 \pmod{23}$. Sends $B = 19$.

Alice computes: $K = 19^6 \pmod{23} = 2$.

Bob computes: $K = 8^{15} \pmod{23} = 2$.

Shared key = **2**.

Fast Modular Exponentiation (repeated squaring)

To compute $a^n \pmod{m}$ efficiently:

Write n in binary. Square-and-multiply:

E.g., $5^6 \pmod{23}$: $6 = 110_2$. $5^1 = 5$, $5^2 = 2$, $5^4 = 4$, $5^6 = 5^4 \cdot 5^2 = 4 \cdot 2 = 8 \pmod{23}$.

Complexity: $O(\log n)$ multiplications.

[HIGH PRIORITY] Advanced Counting

5.1 Recurrence Relations – Recursive Algorithms

Recurrence Relations

A **recurrence relation** defines a sequence a_n in terms of previous values.

Linear homogeneous recurrence of order k :

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

Linear non-homogeneous: Add a function $f(n)$ on the right side.

Solving Linear Homogeneous Recurrences

Step 1: Characteristic equation.

For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$, write: $r^2 = c_1 r + c_2$, i.e., $r^2 - c_1 r - c_2 = 0$.

Step 2: Find roots r_1, r_2 .

Step 3: General solution.

- *Distinct roots:* $a_n = Ar_1^n + Br_2^n$
- *Repeated root $r_1 = r_2 = r$:* $a_n = (A + Bn)r^n$
- *Complex roots $re^{\pm i\theta}$:* $a_n = r^n(A \cos n\theta + B \sin n\theta)$

Step 4: Use initial conditions to find A, B .

Classic Example: Fibonacci Sequence

$$F_n = F_{n-1} + F_{n-2}, F_0 = 0, F_1 = 1.$$

$$\text{Characteristic equation: } r^2 - r - 1 = 0 \Rightarrow r = \frac{1 \pm \sqrt{5}}{2}.$$

$$\text{Let } \phi = \frac{1 + \sqrt{5}}{2} \approx 1.618 \text{ (golden ratio), } \hat{\phi} = \frac{1 - \sqrt{5}}{2}.$$

$$\text{General solution: } F_n = A\phi^n + B\hat{\phi}^n.$$

$$F_0 = 0: A + B = 0 \Rightarrow B = -A. \quad F_1 = 1: A\phi - A\hat{\phi} = 1 \Rightarrow A = \frac{1}{\sqrt{5}}.$$

$$F_n = \frac{1}{\sqrt{5}} (\phi^n - \hat{\phi}^n)$$

Non-Homogeneous Recurrences

For $a_n = c_1 a_{n-1} + c_2 a_{n-2} + f(n)$:

Solution = Homogeneous solution + Particular solution.

Finding particular solution (method of undetermined coefficients):

- $f(n) = d$ (constant): try $a_n^{(p)} = c$ (constant)
- $f(n) = dn^k$: try $a_n^{(p)} = b_k n^k + \cdots + b_0$
- $f(n) = d \cdot s^n$: try $a_n^{(p)} = C \cdot s^n$ (if s is not a characteristic root)
- If s is a characteristic root of multiplicity m : try $a_n^{(p)} = C n^m s^n$

5.2 Recurrence Relations – Divide and Conquer

Divide-and-Conquer Recurrences

Many algorithms split input of size n into a subproblems of size n/b , with $f(n)$ work at each level:

$$T(n) = aT(n/b) + f(n)$$

Examples:

- **Binary Search:** $T(n) = T(n/2) + 1$ ($a = 1, b = 2, f(n) = 1$)
- **Merge Sort:** $T(n) = 2T(n/2) + n$ ($a = 2, b = 2, f(n) = n$)
- **Strassen:** $T(n) = 7T(n/2) + n^2$

5.3 Master's Theorem

Master Theorem — MUST KNOW

For $T(n) = aT(n/b) + f(n)$ with $a \geq 1, b > 1$:

Compare $f(n)$ with $n^{\log_b a}$:

Case 1: $f(n) = O(n^{\log_b a - \epsilon})$ for some $\epsilon > 0$

$$\Rightarrow T(n) = \Theta(n^{\log_b a})$$

(*Recursion dominates — root work negligible*)

Case 2: $f(n) = \Theta(n^{\log_b a} \log^k n)$ for $k \geq 0$

$$\Rightarrow T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$$

(*Both contribute equally*)

Case 3: $f(n) = \Omega(n^{\log_b a + \epsilon})$ and $af(n/b) \leq cf(n)$ for $c < 1$

$$\Rightarrow T(n) = \Theta(f(n))$$

(*Work at top level dominates*)

Master Theorem Examples

Merge Sort: $T(n) = 2T(n/2) + n$.

$a = 2, b = 2 \Rightarrow n^{\log_2 2} = n^1 = n$. $f(n) = n = \Theta(n)$. Case 2 ($k = 0$): $T(n) = \Theta(n \log n)$.

Binary Search: $T(n) = T(n/2) + 1$.

$a = 1, b = 2 \Rightarrow n^{\log_2 1} = n^0 = 1$. $f(n) = 1 = \Theta(1)$. Case 2: $T(n) = \Theta(\log n)$.

Strassen: $T(n) = 7T(n/2) + n^2$.

$n^{\log_2 7} \approx n^{2.807}$. $f(n) = n^2 = O(n^{2.807 - \epsilon})$. Case 1: $T(n) = \Theta(n^{\log_2 7})$.

Growth of Functions – Big-O Hierarchy (Slowest to Fastest)

$$O(1) \subset O(\log n) \subset O(\sqrt{n}) \subset O(n) \subset O(n \log n) \subset O(n^2) \subset O(n^3) \subset O(2^n) \subset O(n!)$$

Key definitions:

- $f = O(g)$: f grows no faster than g (upper bound)
- $f = \Omega(g)$: f grows no slower than g (lower bound)

- $f = \Theta(g)$: f grows at the same rate as g (tight bound)

5.3 Generating Functions

Generating Functions – Definition

The **ordinary generating function (OGF)** of a sequence $\{a_n\}_{n=0}^{\infty}$ is the formal power series:

$$G(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

We treat x as a formal variable (convergence is not the concern).

Exponential generating function (EGF):

$$\hat{G}(x) = \sum_{n=0}^{\infty} a_n \frac{x^n}{n!}$$

(Used for permutations and labeled structures.)

Standard Generating Functions — Memorise!

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots$$

$$\frac{1}{1-ax} = \sum_{n=0}^{\infty} a^n x^n$$

$$\frac{1}{(1-x)^k} = \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} x^n$$

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k \quad (\text{Binomial theorem})$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad (\text{EGF for } a_n = 1)$$

Key trick: $x \cdot G'(x) = \sum_{n=1}^{\infty} n a_{n-1} x^n$ (differentiating shifts and multiplies by n).

Example: Generating Function for $a_n = 2^n$

$$G(x) = \sum_{n=0}^{\infty} 2^n x^n = \sum_{n=0}^{\infty} (2x)^n = \frac{1}{1-2x} \text{ for } |x| < 1/2.$$

So the sequence $1, 2, 4, 8, 16, \dots$ has OGF $\frac{1}{1-2x}$.

5.4 Applications of Generating Functions in Counting

Counting with Generating Functions

Key Idea: To count selections from sets, multiply their generating functions.

Selecting r items with repetition from n types: The number of solutions to $x_1 + x_2 + \dots + x_k = n$ with $0 \leq x_i \leq M$ is the coefficient of x^n in $\left(\sum_{j=0}^M x^j\right)^k = \left(\frac{1-x^{M+1}}{1-x}\right)^k$.

Unlimited repetition: coefficient of x^n in $\frac{1}{(1-x)^k} = \binom{n+k-1}{k-1}$.

Example: Distributing Items

Problem: In how many ways can we choose r items from 3 types with unlimited repetition?

$$G(x) = \frac{1}{1-x} \cdot \frac{1}{1-x} \cdot \frac{1}{1-x} = \frac{1}{(1-x)^3}.$$

Coefficient of x^r is $\binom{r+2}{2}$.

Problem: Ways to make change for n cents using 1-cent, 5-cent, 10-cent coins: $G(x) = \frac{1}{1-x} \cdot \frac{1}{1-x^5} \cdot \frac{1}{1-x^{10}}$. Coefficient of x^n gives the count.

5.5 Generating Functions for Solving Recurrence Relations

Method: Solving Recurrences via GFs

Steps:

1. Multiply both sides of recurrence by x^n and sum over n .
2. Express in terms of $G(x) = \sum a_n x^n$ and known GFs.
3. Solve for $G(x)$ as a closed-form rational function.
4. Expand using partial fractions to find a_n .

Worked Example: Recurrence $a_n = 3$ times a_{n-1}

Let $G(x) = \sum_{n=0}^{\infty} a_n x^n$.

Recurrence: $a_n - 3a_{n-1} = 0$ for $n \geq 1$.

$$\sum_{n=1}^{\infty} a_n x^n - 3x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} = 0$$

$$(G(x) - a_0) - 3xG(x) = 0$$

$$G(x)(1-3x) = a_0 \Rightarrow G(x) = \frac{a_0}{1-3x} = \sum_{n=0}^{\infty} a_0 3^n x^n$$

Therefore: $\boxed{a_n = 2 \cdot 3^n}$.

Worked Example: Fibonacci via GF

$F_n = F_{n-1} + F_{n-2}$, $F_0 = 0$, $F_1 = 1$.

$$G - xG - x^2G = F_0 + (F_1 - F_0)x = x$$

$$G(x) = \frac{x}{1-x-x^2}.$$

$$\text{Partial fractions: } \frac{x}{1-x-x^2} = \frac{1/\sqrt{5}}{1-\phi x} - \frac{1/\sqrt{5}}{1-\hat{\phi}x}$$

$$\Rightarrow F_n = \frac{\phi^n - \hat{\phi}^n}{\sqrt{5}} \text{ (same closed form as before).}$$

Using Partial Fractions in GF (General Technique)

If $G(x) = \frac{P(x)}{(1 - r_1x)(1 - r_2x)}$ with $r_1 \neq r_2$:

Write $G(x) = \frac{A}{1 - r_1x} + \frac{B}{1 - r_2x}$

Then $a_n = Ar_1^n + Br_2^n$ — exactly the characteristic root formula!

If repeated root: $G(x) = \frac{A}{1 - rx} + \frac{B}{(1 - rx)^2}$ gives $a_n = (A + Bn)r^n$.

Quick Reference Summary Sheet

Abstract Algebra Cheat Sheet

Group: CAII (Closure, Assoc, Identity, Inverse)

Abelian: + commutativity

Ring: abelian group under +; assoc $\times$; distributive

Field: ring where nonzero elements $\times$ -invertible

Lagrange: $|H| \mid |G|$

Corollaries: $\text{ord}(a) \mid |G|$; $a^{|G|} = e$

Cyclic generator: $\text{gcd}(a, n) = 1$ generates $\mathbb{Z}_n$

generators: $\phi(n)$

ord(g^k) = $n / \text{gcd}(n, k)$

Primitive root mod p : g^k cycles through $\{1, \dots, p-1\}$

Fermat: $a^{p-1} \equiv 1 \pmod{p}$

DH Key Exchange: $K = g^{ab} \pmod{p}$

Advanced Counting Cheat Sheet

Recurrence (homogeneous): Char. eqn $\rightarrow$ roots $\rightarrow a_n = Ar_1^n + Br_2^n$ (use IC)

Master Theorem: Compare $f(n)$ vs $n^{\log_b a}$:

Case 1 (f smaller): $T = \Theta(n^{\log_b a})$

Case 2 (f equal $\times \log^k$): $T = \Theta(n^{\log_b a} \log^{k+1} n)$

Case 3 (f larger, regularity): $T = \Theta(f(n))$

GF shortcuts: $\frac{1}{1-x} = \sum x^n$; $\frac{1}{1-ax} = \sum a^n x^n$; $\frac{1}{(1-x)^k} = \sum \binom{n+k-1}{k-1} x^n$

Recurrence via GF: Multiply by x^n , sum, express $G(x)$, partial fractions $\rightarrow a_n$.

Modular Arithmetic Quick Reference

$\phi(p) = p - 1$ for prime p

$\phi(p^k) = p^{k-1}(p - 1)$

$\phi(mn) = \phi(m)\phi(n)$ if $\text{gcd}(m, n) = 1$

$a^{-1} \pmod{n}$: use extended Euclidean

$a^{\phi(n)} \equiv 1 \pmod{n}$ (Euler, $\text{gcd}(a, n) = 1$)

$a^{p-1} \equiv 1 \pmod{p}$ (Fermat)

Fast exponentiation: $O(\log n)$ squarings

Last-Minute Exam Tips

1. For **Lagrange**: always check $|H| \mid |G|$ first to verify it's a valid subgroup.
2. For **cyclic group generators**: compute $\text{gcd}(a, n)$ — equal to 1 means generator.
3. For **Diffie-Hellman**: remember Alice and Bob each compute the *other's* public key raised to their own private key.
4. For **Master Theorem**: always compute $\log_b a$ first, then compare with $f(n)$.
5. For **GF recurrences**: multiplying through by x^n and summing is the key first step.
6. For **proof by induction**: always state base case AND inductive step clearly.
7. For **fields vs rings**: fields require multiplicative inverses for all nonzero elements.